



Duyfken Deck Training

Version 2.1 : 2016

Name: _____

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SAFE LINE HANDLING

Safe line handling is fundamental. We don't want any injuries aboard.

Most control lines on *Duyfken* can safely be handled alone, barring tacks. Nevertheless, always ask for assistance if you have any doubt, or are struggling.

Never have your hands within 40cm from a block or fair-lead. You run the risk of serious damage.

Never stand hand in a Bight of rope or take a turn around your hand.

Always have both feet on deck when sweating or tailing a line.

No locking hitches are to be used aboard *Duyfken*. Natural fibre will jam when wet.

Every line has a designated Belaying Pin or Cleat. Ensure they are all belayed and coiled after each sail evolution.

Coil all lines clockwise from the standing end neatly to allow them to run safely.

Treat all lines as if they are under heavy load, leaving one turn on the cleat or pin.

Always haul and ease 'hand over hand' so as to maintain control of the line.

To belay or make fast any line, the rule 'clockwise around the top' of any vertical pin or cleat should be observed, ensuring a fair lead.

NOMENCLATURE

Avast	To cease hauling/easing but to keep control of your line.
Belay / Make-fast	To secure control lines to a belaying pin or cleat.
Bitter end	The free end of any line, terminated with a whipping.
Come-up	To drop the line being hauled on so it can be belayed, even casting it toward the cleat.
Coil down	To coil away the gear, and stow it on pins or cleats.
Ease away	To ease or pay out slack, hand over hand.
Ease to the stopper	To ease hand over hand until the weight is taken by a stopper, the command to 'come up' will follow (used on halyards).
Furl	To stow a sail after use.
Haul away	To haul upon a line, hand over hand.

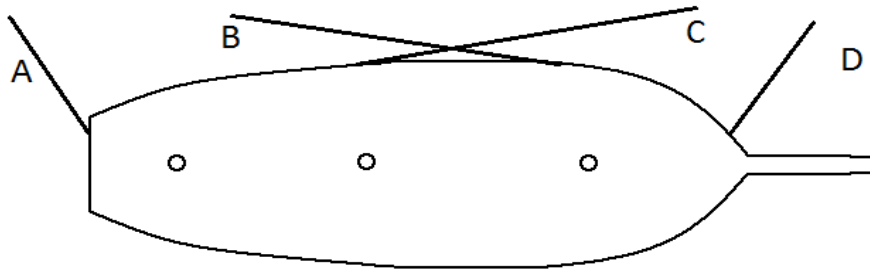
Leeward	The ' Lee ' side of the ship is the direction the wind is heading, or the direction downwind from the point of reference.
Let-go / Cast-off	To remove the belay, allowing the line to pay out uncontrolled.
Loose	To remove the gasket stowing the sail, prior to setting.
Overhaul	To put excess slack in a line, to allow free movement.
Port/Starboard Tack	A ship is said to be 'on a port tack' when the port yardarm is forward, and the port tack is hauled taut, 'a starboard tack' is when the starboard yardarm is forward and the starboard tack is taut.
Stand-by	Prepare a particular line for handling, placing the coil on deck, and taking the belay down to one turn.
Standing end	This is the working end of any line, the end attached to something.
Tacking	To put the bow through the eye of the wind.
Wearing	To put the stern through the eye of the wind.
Windward	The ' Weather ' side of the ship, where the wind is coming from, or the direction upwind from the point of reference.

MOORING STATIONS

When mooring *Duyfken*, all mooring stations need to be manned.

ALL Mooring lines need a heaving line bent on well in advance and coiled neatly.

- | | |
|--------------------|---|
| Sternline 'A' | - Belayed in the tiller-flat on the Panama Cleat. Communication is difficult here and the helmsman MUST relay any orders from Master to tiller-flat, heaving line thrown from poop deck. |
| Forward Spring 'B' | - Belayed to the forward most Cavel cleat (same location the <i>Main Tack</i> is belayed). Heaving line thrown from Midships. |
| Aft Spring 'C' | - Belayed on the AFT most Cavel cleat (same location the <i>Fore Sheet</i> is belayed). Heaving line thrown from Midships (usually the first line ashore). |
| Headline 'D' | - Belayed on the windlass bit (same location the <i>Fore Tack</i> is belayed). Heaving line thrown from beak or fore-deck. |



- | | |
|---------------------|--|
| Roving fender 1 & 2 | - Small roving fenders are on stand-by during berthing. DO NOT take a turn around your hand. The lines from the fender Must be held in a manner that allows you to let go if needed. |
| Bow Lookout | - Standing on the fore-deck keeping a lookout, if asked, calling distance between the ship and any other obstacle ahead and abreast. |
| Helmsman | - The Master will be conning the ship and calling helm commands in degrees for manoeuvring in close quarters. 5 degrees 10 degrees and 'Hard' to Port or Starboard. |
| Botte-loef | - The botte-loef must be run out after leaving the berth and run in before mooring. |

HELM INSTRUCTIONS

Always repeat command given - e.g. Master asks “5 to starboard”, the helmsman must repeat “5 to starboard aye”. This is essential for safe manoeuvring in port and at sea.

When sailing by compass, the Master will give you a compass heading to follow.

When sailing 'Full and by', you are to sail as close to the wind as possible, keeping the sail full. In addition, the helmsman must:

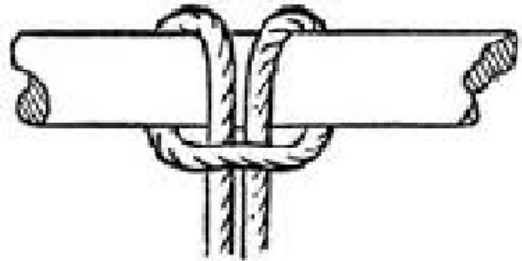
1. remain standing at all times.
2. keep two hands on the whipstaff.
3. not use feet to control the whipstaff; moving off the steps instead to keep control of the whipstaff.
4. not be spoken to except for necessary helm orders.
5. not to eat or drink.
6. complete tricks of at least one hour's duration.
7. relay **ALL** commands for the stern-line, when berthing.
8. not be relieved without the permission of the **OOW**.

USEFUL KNOTS

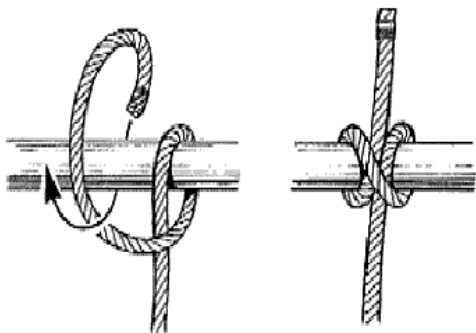
Rolling hitch



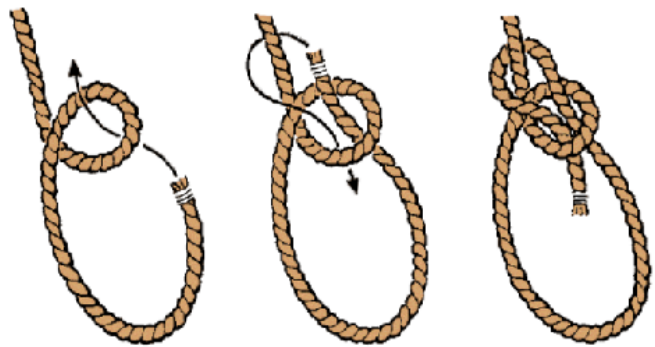
Cow hitch



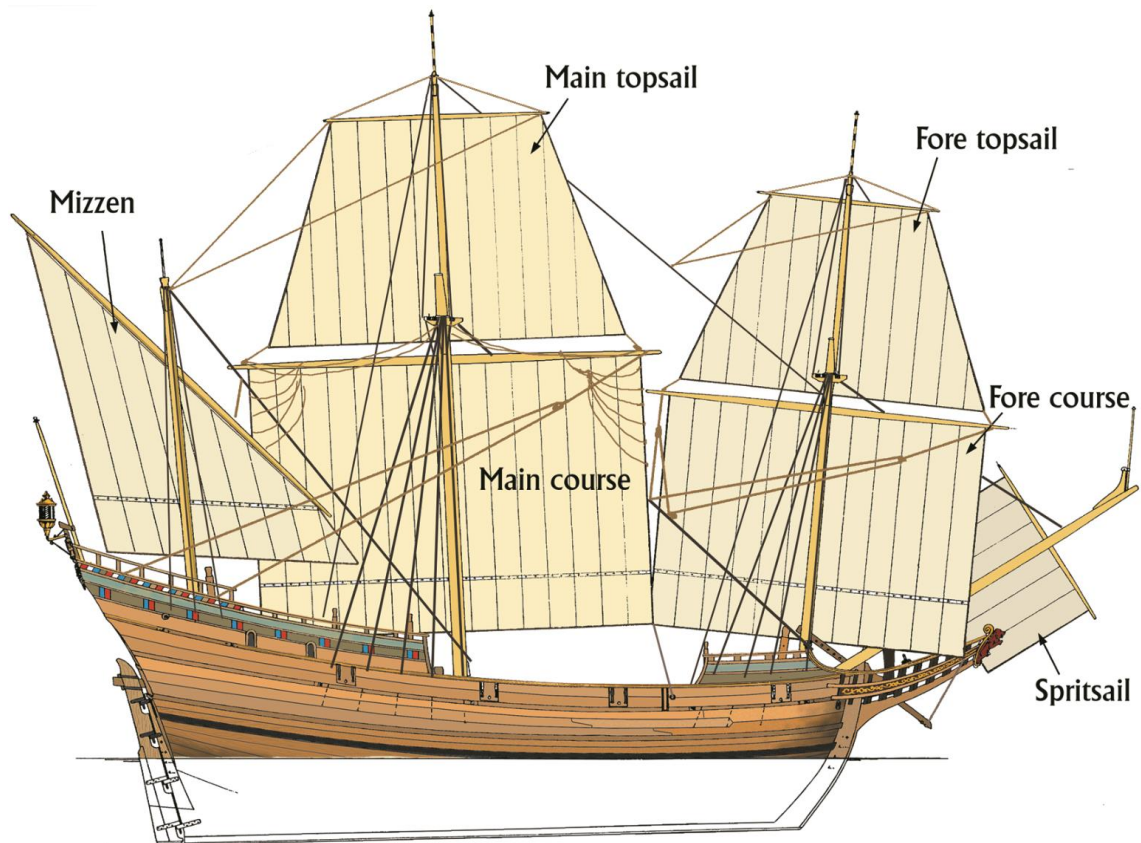
Clove hitch



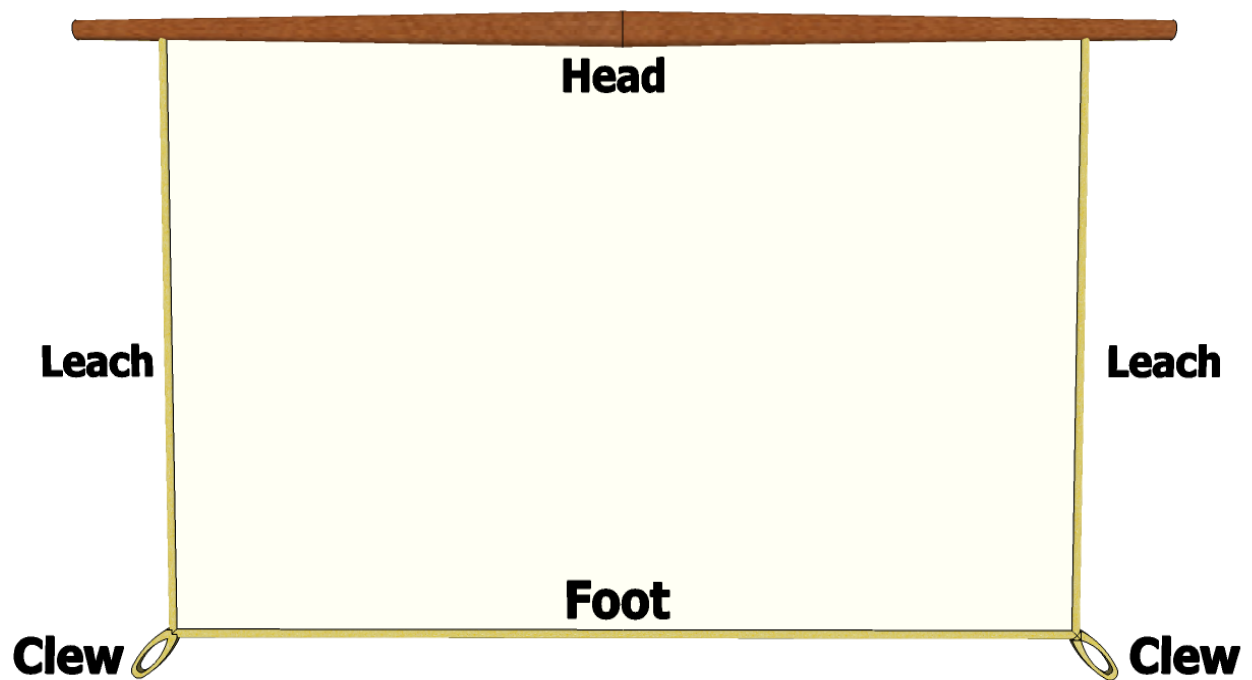
Bowline



SAIL PLAN

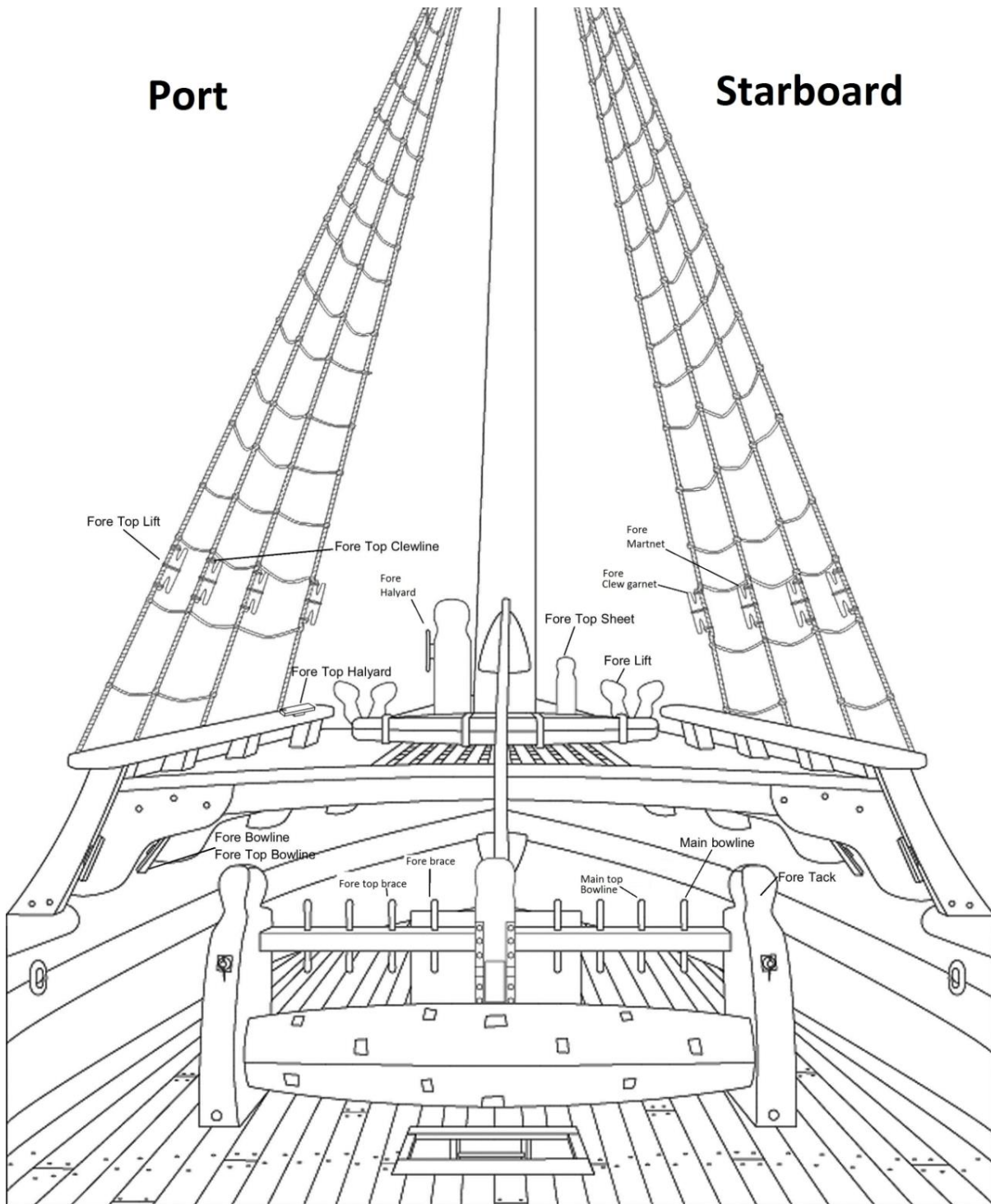


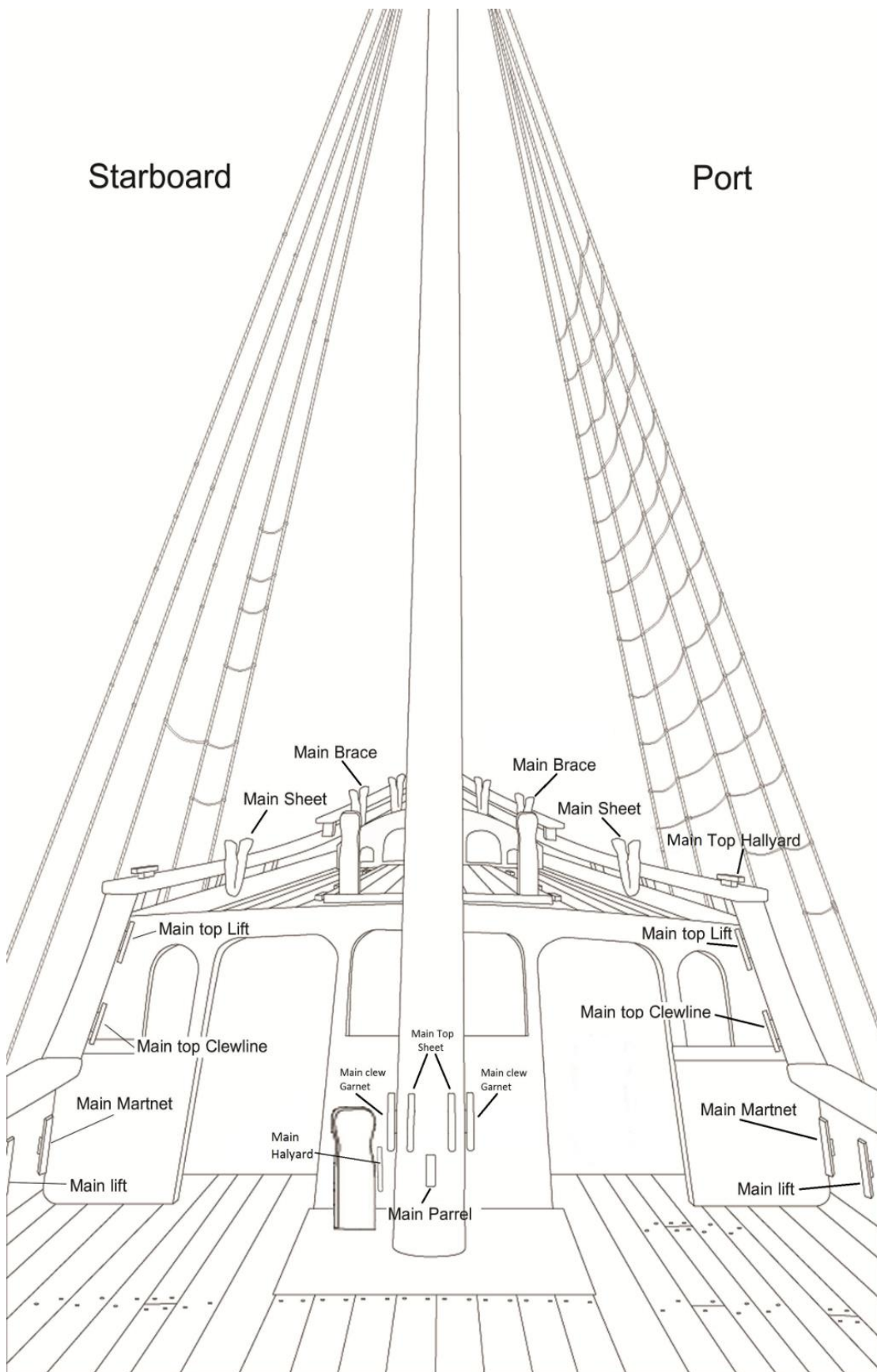
PARTS OF A SQUARE SAIL



LINE LOCATIONS AND USES

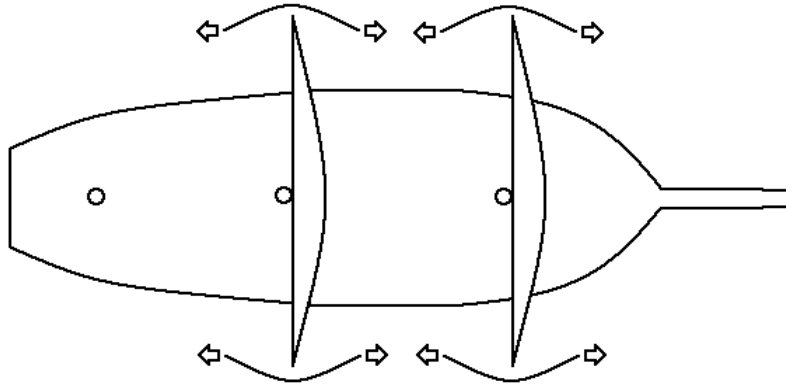
As the fore deck is cluttered, some lines have been omitted. It can be assumed that lines are belayed symmetrically. e.g. the tops'l lift will be belayed on the aftmost fore shroud on both port and starboard.





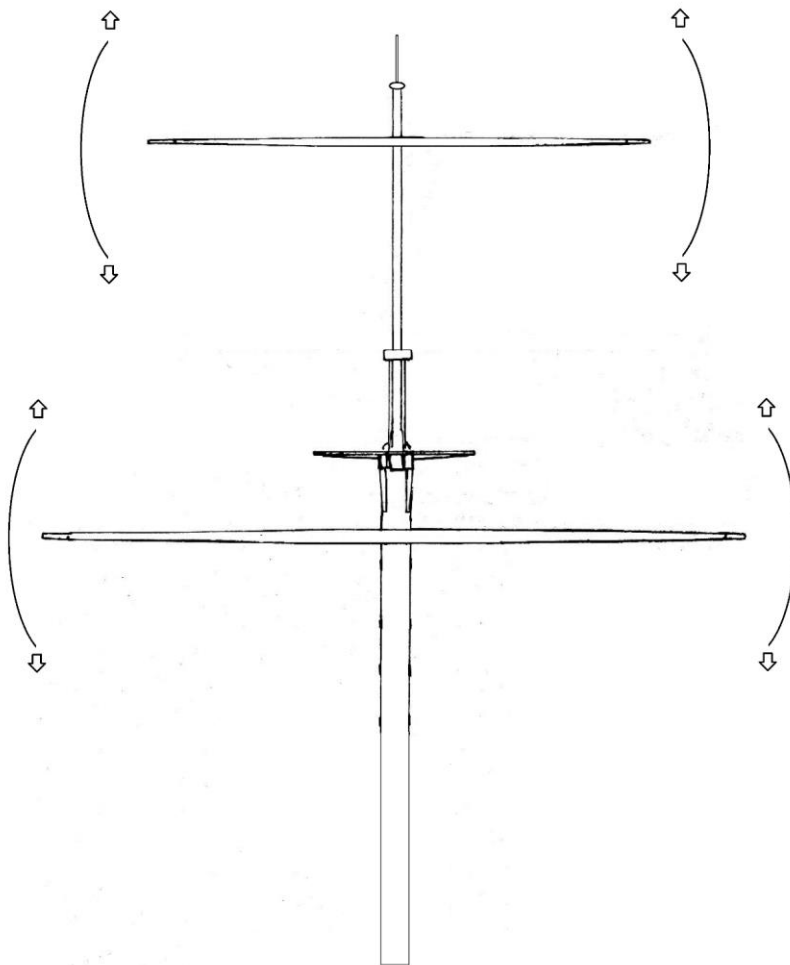
BRACES

Braces control the orientation of the yard in the horizontal plane. The port and starboard brace need to be tended together, one eased while the other is hauled.



LIFTS

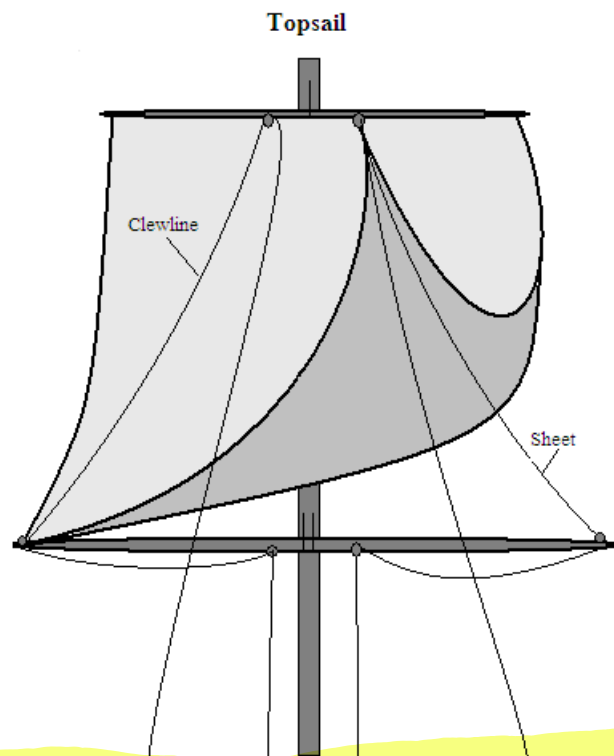
Lifts control the orientation of the yard in the vertical plane. The port and starboard lifts need to be tended together, one eased while the other is hauled.



SHEETS, TACKS & CLEWLINES/CLEW-GARNETS

Sheets control the lower corners or 'Clews' of a square sail.

Topsail sheets stretch the lower corners of the topsail out to the end of the Course yardarms and are opposed by the Clewlines.



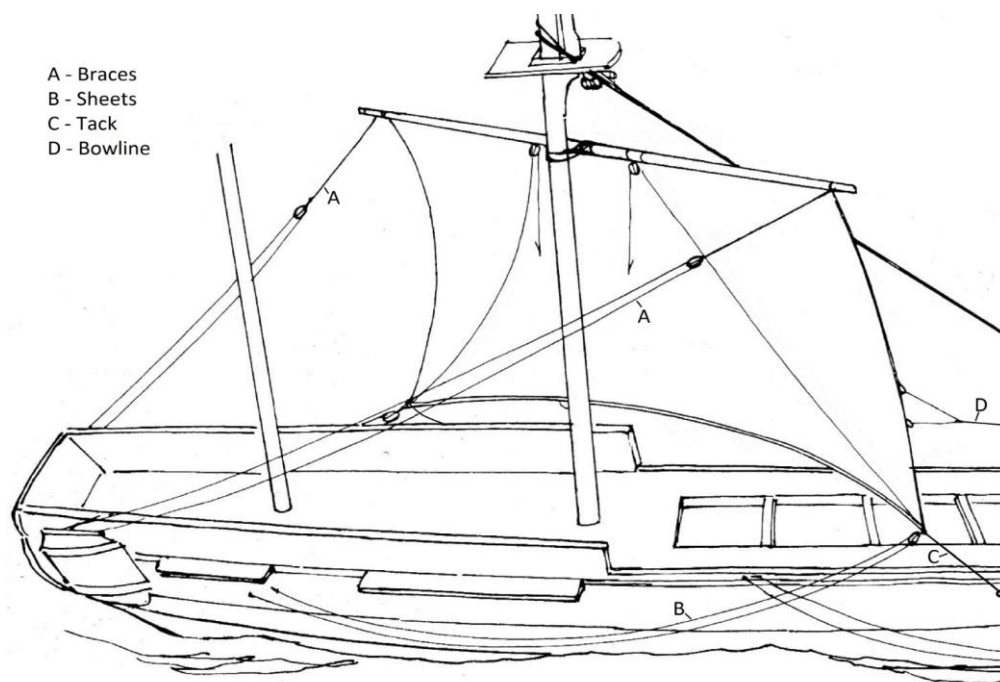
Page Break

Course Sheets 'B' control the 'Loose' clews of the course, pulling them aft.

Tacks 'C' oppose the Sheets 'B', pulling the clew forward.

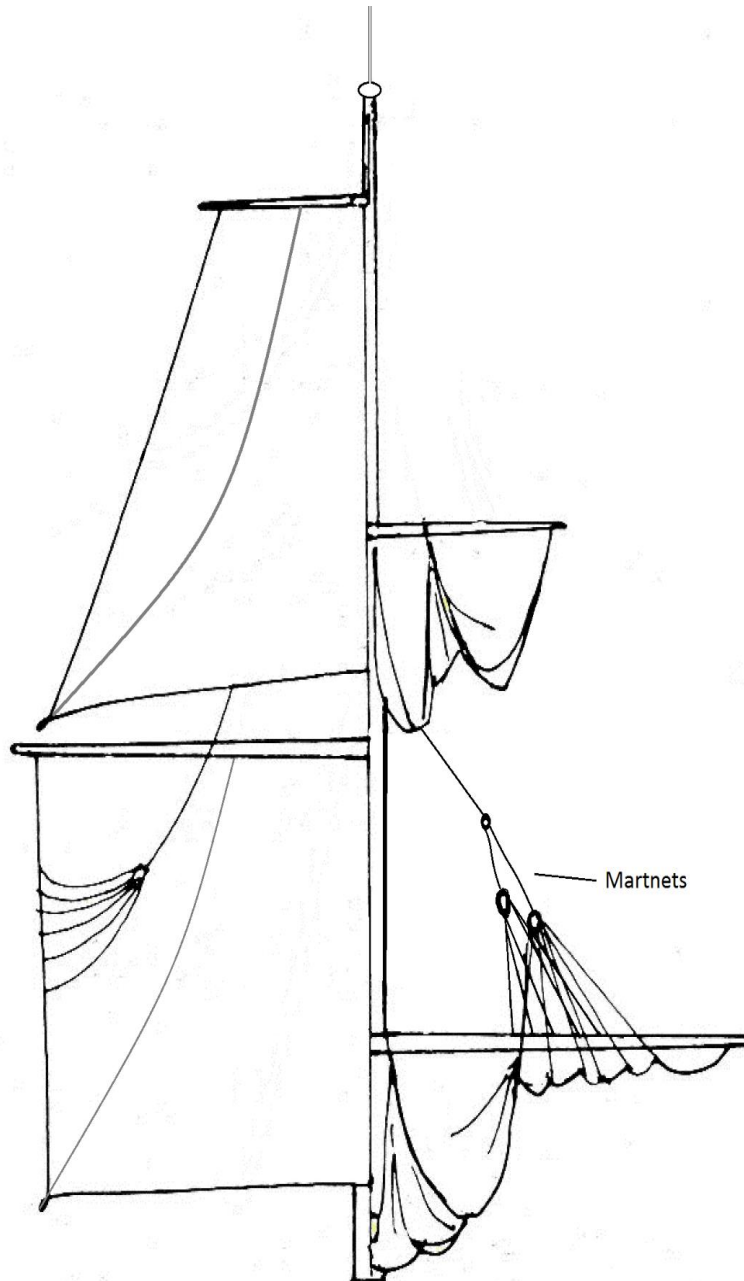
When 'Bracing for a Port Tack', the Starboard Brace, Starboard Sheet and Port Tack need to be hauled and the Port Brace and Port Sheet and Starboard Tack must be eased.

Bowlines 'D' Haul the weather leach of a sail forward to allow the sail to fill reliably when hard on the wind.



MARTNETS

Martnets are used to contain the sail outboard of the vessel when furling, to eliminate the need to send crew outboard of the bulwarks. Martnets are hauled taut once the course yard has been lowered to deck.



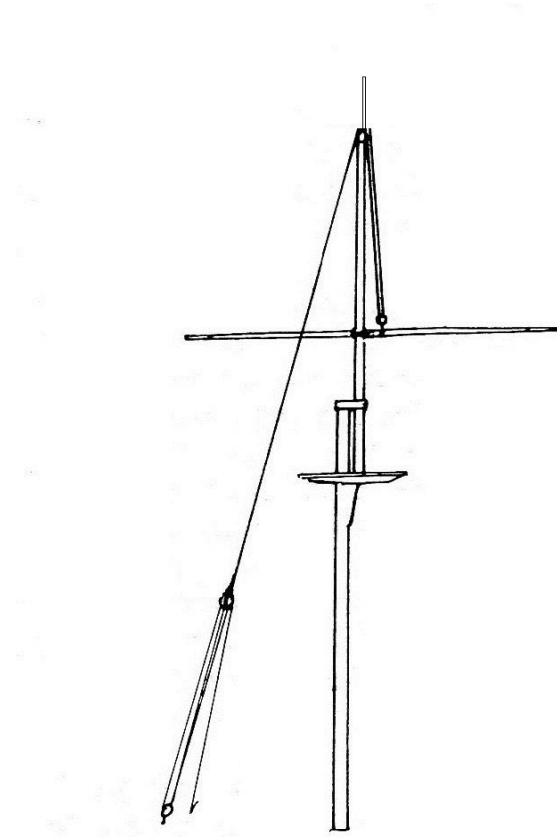
This Image shows the Course and Topsails set (Left) and 'in their gear' ready to be furled (Right)

HALYARDS

Halyards as the name suggests - haul the yards aloft.

The Course yards are approximately 450kg without canvas and must be treated with care. 6 people minimum hauling hand over hand to ensure no loss of progress. The halyard then has a stopper applied and held whilst the line is belayed.

Tops'l halyards require 3-4 hands and **Must** be 'Stoppered' or 'Married' by one crew member and belayed by another.



PROCEDURE FOR SETTING SAIL

Course

Lowering Course to loose sail

- **Cast-off** Bowlines
- **Cast-off** Tops'l Sheets
- Set Lifts to their leathers
- **Tend** Braces
- **Tend** Parrel if rolling
- **Ease** Halyard
- **Ease** Lifts
- **Cast-off** gasket and coil around halyard tye

Setting Courses

- **Cast off** Bowlines
- Remove rolling tackle if rigged
- **Cast off** Parrel and **Tend** if ship is rolling
- **Board the** Tack if hard on the wind
- **Tend** the Braces
- **Haul** the Halyard
- **Tend** the Sheets
- **Tend** Clewlines
- Sheet home
- Belay all & coil down

Dousing Courses

- **Cast-off** Bowlines
- Square Course Yard
- Set lifts to their leathers
- **Cast-off** Martnets
- **Tend** Parrel if needed
- **Tend** Braces
- **Cast-off** Tops'l Sheets
- **Ease** Halyard until yard is taken in her Lifts
- **Haul** Clewlines (*weather clew first*)
- Furl sail
- **Haul** Halyard
- Take slack out of Lifts and Sheets
- **Belay** all & coil down

Topsail

Setting, Assuming the Course yard is mast-headed and tops'l is loosed

- **Cast-off** Bowlines
- **Let go** Tops'l clewlines
- **Haul** Tops'l Sheets home
- **Tend** the Tops'l Braces
- **Haul** Tops'l Halyard
- Belay all & coil down, leave slack in clewlines

Dousing

- **Cast off** Bowlines
- Set Tops'l Lifts to their leathers
- **Square** Tops'l Yard
- **Ease** Tops'l Halyard, **Haul** Tops'l Clewline until yard is in her Lifts
- **Ease** Tops'l Sheet, **Haul** Tops'l Clewline (weather clew first)
- Belay all & coil down
- Furl sail

Mizzen

Lowering to Loose

- Bring in flag if flying & cast off flag halyard
- **Cast-off** Topping lift
- **Tend** Bowlines, Keeping yard peak away from ensign staff, lantern and radar/EPIRB
- **Ease** Halyard until the yard sits on the rail
- **Cast-off** gasket and coil around halyard tye

Raising and setting

- **Tend** Bowline
- One hand to Manhandle Knock (Lower end of yard)
- **Tend** Topping Lift
- **Tend** Parrel
- **Haul** Halyard
- **Haul** taut Parrel
- **Sheet** home
- Belay all & coil down

Dousing

- **Cast-off** Topping Lift
- **Haul** Clewline
- **Ease** sheet
- **Tend** Bowlines, Keeping yard peak away from ensign staff, lantern and radar/EPIRB
- **Ease** Halyard until the yard sits on the rail
- Furl sail
- Belay and Coil

Spritsail

Lowering to loose

- **Cast-off** Lifts
- **Ease** Halyard until the yard is in its braces
- **Haul** Clewlines
- **Ease** Braces
- Cast-off Gaskets, coil around Halyard tye

Setting

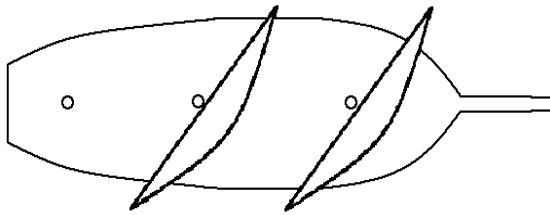
- **Haul** Braces
- **Haul** Halyard
- **Tend** Lifts
- **Tend** Sheets
- **Ease** Clewlines
- **Haul** Sheets
- **Haul** taut Lifts
- **Belay** all coil

Dousing

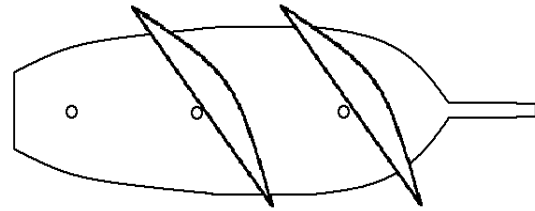
- **Cast-off** Lifts
- **Ease** Halyard until the yard is in its braces
- **Haul** Clewlines
- Clear sail from figurehead
- **Ease** Braces
- Furl sail
- Stow in beak until moored
- Belay & coil

BASIC SHIP HANDLING

Ship on a Port Tack

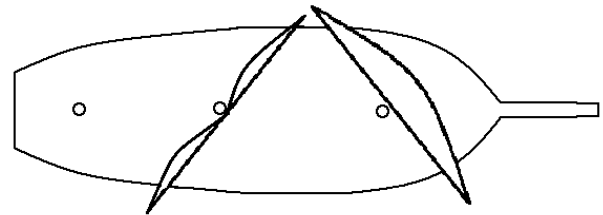
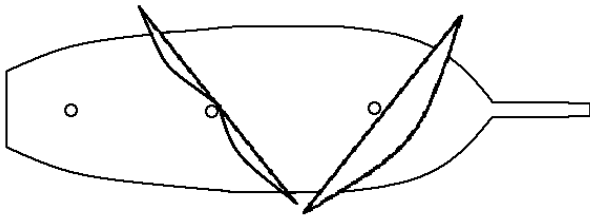


Ship on a Starboard Tack

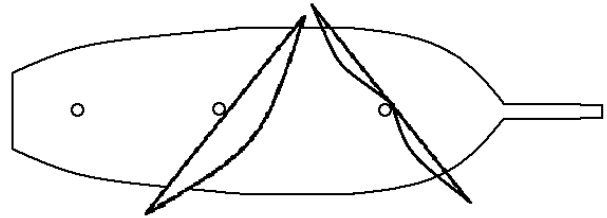
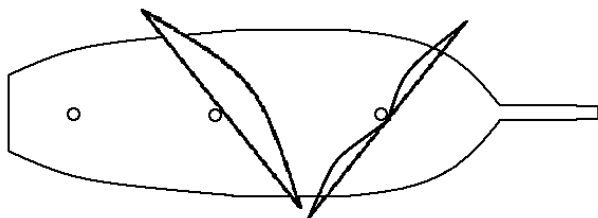


Heaving to is the quickest way to stop a ship at sea. Ideal for a Man Over Board incident.

A Ship 'Hove to' with her Mainmast aback, on either tack.



A Ship 'Hove to' with her Foremast aback, on either tack.



ADVANCED SHIP HANDLING

Spritsail – balances the mizzen. Old paintings seldom show the spritsail and mizzen set. While we too have found that the ship handles very well in most conditions under just courses and topsails, the spritsail is still a useful sail on longer passages. Have the sail well dipped when going to windward; lee sheet hauled well aft and weather sheet on just enough for the sail to draw. When braced up the sail sets like a “settee” and pulls well. Let go the lee sheet when tacking and then leave sail until last; simply haul around the braces and trim the sheets.

Mizzen – is needed for the ship to tack without the topsails set and is useful for balance when going to windward in steep head seas when the ships’ head is being knocked to leeward. When dipping the yard, have plenty of hands on the job, the risk of a MOB being fairly high in a seaway. When making short tacks, it is not necessary to dip the yard each time; the sail is still useful when set to weather.

Maincourse – when setting, tend sheets carefully to avoid clews and sheets going into water and fouling propellers. Tacks can be used to haul the clews inboard. If you have the time and the searoom, have Maincourse furled and lines squared away before engaging props. Keep yard square when hoisting and lowering, otherwise it fouls on the shrouds.

If there is difficulty getting tack down, ease lee brace and sheet; hauling aft again once the tack is boarded. Keep lifts and parrel falls well slack when bracing sharp up, yard will come right around onto the backstay.

When hoisting or lowering yard, have parrel fall tended to prevent yard slatting about too much when the vessel is rolling. Bonnet remains on sail until wind speed well up over 30 knots and has stood with no problems in 40 knots. Maincourse usually first to lift when getting too close to the wind. Leather on lifts and martinets to mark lowered position. Tack to the chesstree in ship’s side, in flat water, tack may be taken to windlass to allow vessel to stand closer to wind. This has little benefit at sea as it just increases leeway.

Forecourse – with bonnet on - get the weather clew well down past the beak. Lifts need to be slack ensuring the yard is cockbilled down to weather when going to windward. Ease sheet when tacking but ensure sheet is then tended to prevent excessive flogging. Ensure 1 or 2 hands in beak to handle tacks, getting weather tack well down and overhauling lee tack.

Topsails – lower yards to about half hoist when wind is up over 20-25 knots to avoid excessive bending of topmasts. Halyard can be shifted to weather to provide extra support if on one tack for an extended period of time. In any case, halyards should be shifted from time-to-time to provide even wear on both sides. Bowlines need to be well hauled out when going to windward, especially at half hoist. Steady taut the lifts in sloppy conditions. Leather on lifts to mark lowered position.

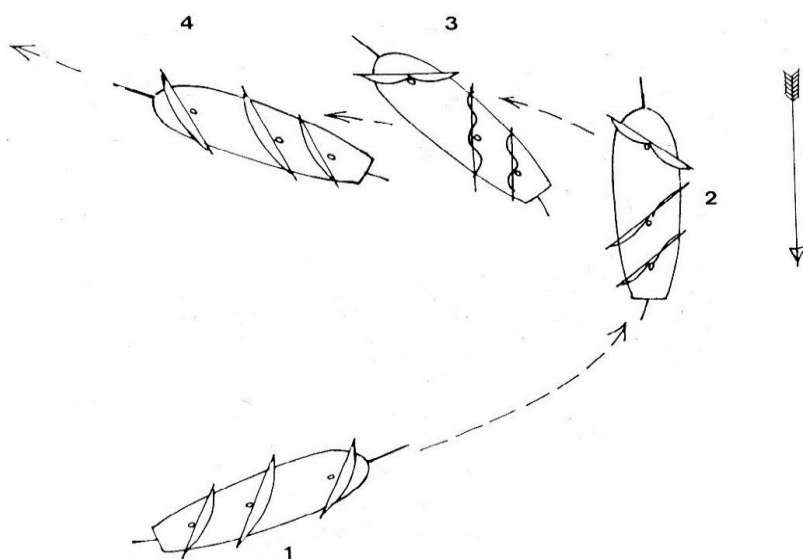
Sail combinations – when coming to anchor under sail, try and do so under topsails if possible as it is difficult to work the forecourse and the anchors together. Topsails and forecourse is a good rig for daysails, although, without the main, she can be difficult to tack (use the mizzen). Both courses and both topsails is the most common rig in moderate conditions. She is sluggish under courses alone when on the wind.

Ship heaves to nicely under either fores'l or mains'l, although under the fore the helm is needed to be hard over to prevent her turning off the wind too far.

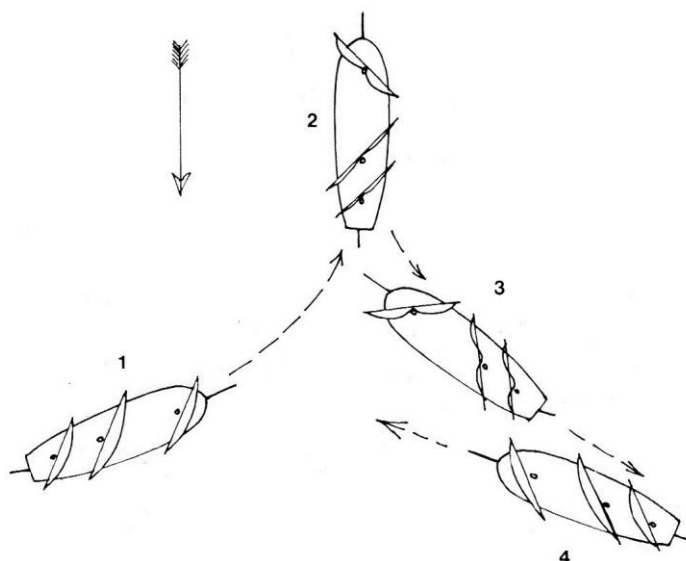
Sail combinations (continue..)

Under the main, do not haul the sheet too far aft to prevent her turning up into the wind. When running downwind, she goes nicely under the main squares – steering is still very easy and it is a much bigger sail area than the fore squares. **Scudding in** heavy weather is usually under the fores'l. In heavier weather, the 4 lower sails are a good rig, the spritsail and mizzen going some way to making up for the loss of sail area of the topsails and before you need to shorten right down to just the courses.

Tacking and wearing - The ship tacks easily under courses and topsails. Ensure you let go the foresheet and haul the main sheet well aft and be prepared to “brace to”, checking the foreyards in just a little to ensure they come aback and push the head through the wind. Do not haul the main yards too early or she will stop. The mizzen is needed to tack if the topsails are not set.



Making Sternboard while tacking - If the winds are light or the ship has lost her headway, she will make sternway. In moderate conditions a successful change of tack can still be made if the helm is put over to the opposite side (see rudder change from fig 1 to fig 3 below).



Wearing - As wearing is generally fairly slow, make sure you have adequate searoom. Hand the mizzen completely and square the main yards early – keep them lifting until well off the wind but ensure they do not come aback.

